Train

x_1	x_2	y
1	1	2
3	1	4

Test

$$x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$h_\theta(x) = \theta^T \phi(x) \quad , \quad \theta \Rightarrow \mathbb{R}^p$$

$$\phi(x) \Rightarrow \mathbb{R}^p$$

$$\boxed{h_\theta(x) = \sum_{i=1}^{n} \beta_i K(x^{(i)}, x)}$$

Let us assume :

$\beta_1 = 1$

$\beta_2 = 1$

Kernel = Radial Basis Function (RBF)

$$K(x, z) = \exp(-\|x - z\|^2)$$

where

$$\|x - z\|^2 = (x_1 - z_1)^2 + (x_2 - z_2)^2$$

$$h_\theta(x) = \sum_{i=1}^{n} \beta_i K(x^{(i)}, x)$$

$$= \beta_1 K(x^{(1)}, x) + \beta_2 K(x^{(2)}, x)$$

$= \beta_1 \exp(-\|x^{(1)} - x\|^2) + \beta_2 \exp(-\|x^{(2)} - x\|^2)$

i) $\|x^{(1)} - x\|^2$

$= (x_1^{(1)} - x_1)^2 + (x_2^{(1)} - x_2)^2$

$= (1-2)^2 + (1-1)^2 = 1$

ii) $\|x^{(2)} - x\|^2$

$= 1.$

$K(x^{(1)}, x) = \exp(-\|x^{(1)} - x\|^2)$

$= \exp(-1)$

$= 0.367$

$K(x^{(2)}, x) = 0.367$

$h_\theta(x) = \beta_1 K(x^{(1)}, x) + \beta_2 K(x^{(2)}, x)$

$= 1 \times 0.367 + 1 \times 0.367$

$\boxed{h_\theta(x) = 0.734}$